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System and method for operating utility vehicles

Abstract

A system for operating a driverless utility vehicle in a restricted area, including: a transfer module for transferring control of the utility vehicle at an entrance: from the driver of the utility vehicle to the system, and at an exit: from the system to the driver of the utility vehicle, wherein the restricted area has the entrance, the exit and a predetermined target point; a movement module for autonomously moving the utility vehicle from the entrance to the predetermined target point and from the predetermined target point to the exit; and a performance module for autonomously performing an action on the utility vehicle while the utility vehicle is situated at the predetermined target point, so as to change a state of the utility vehicle by the action. Also described are a related management system, a method, and a computer readable medium.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6278917	12/2000	Bauer et al.	N/A	N/A
11300969	12/2021	Balogh	N/A	G08G 1/148
2004/0267411	12/2003	Mayer et al.	N/A	N/A
2015/0151769	12/2014	DeSanzo	701/1	G08G 1/133
2016/0334797	12/2015	Ross et al.	N/A	N/A
2017/0351267	12/2016	Mielenz	N/A	N/A
2018/0061237	12/2017	Erickson et al.	N/A	N/A
2019/0163192	12/2018	Takeda	N/A	B60P 1/04
2020/0242924	12/2019	Publicover	N/A	G08G 1/087

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
104044629	12/2013	CN	N/A
104837705	12/2014	CN	N/A
10322765	12/2004	DE	N/A
102008036660	12/2008	DE	N/A
102012021282	12/2013	DE	N/A
102014011796	12/2015	DE	N/A
102014221777	12/2015	DE	N/A
102014224113	12/2015	DE	N/A
102014224124	12/2015	DE	N/A
1480097	12/2003	EP	N/A
2136275	12/2008	EP	N/A
2910453	12/2014	EP	N/A

OTHER PUBLICATIONS

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Background/Summary

RELATED APPLICATION INFORMATION (1) The present application is a continuation application of U.S. patent application Ser. No. 16/329,955 filed Mar. 1, 2019, which is a U.S. National Phase application of International Patent Application PCT/EP2017/070641 filed on Aug. 14, 2017, all of which claim priority to and the benefit of German patent application no. 10 2016 116 857.2, which was filed in Germany on Sep. 8, 2016, the disclosures of all of which are incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present invention relates to a system and to a method for operating driverless utility vehicles, and in particular to autonomous operation of utility vehicles in a restricted area.

BACKGROUND INFORMATION

(2) In the use of utility vehicles, transport time and the utilization of the driver's working hours are significant factors for the effectiveness of a vehicle fleet. The majority of transport typically takes place on public roads with standardized rules, which often do not place any high requirements on the driver. Problems may however occur when the utility vehicle is used in specific areas where the surroundings are unique, space is limited and specific maneuvers need to be executed that often place high requirements on the driver. Such specific areas are for example loading stations or goods yards that often do not provide enough space and yet require the driver to maneuver in various directions.

(3) A lot of time is then lost in particular when the driver has to execute specific, non-standardized maneuvers within areas with which he is not familiar and that pose particular difficulties. Especially because the driver often feels uncertain in such situations, the likelihood of accidents is increased. By way of example, accidents frequently occur when the driver has to reverse a utility vehicle into a tight area in a complex environment.

(4) Even though external interfering influences are fairly low in such environments/areas and the speed of the utility vehicle is fairly slow, such systems only offer insufficient solutions in this respect. DE 10 2014 221 777 A1 thus discusses a device that parks a vehicle at a specific position in a parking space and turns it off on the basis of a digital map. DE 10 2014 011 796 A1 discusses a further system for the autonomous navigation of vehicles, in which the vehicle is moved from a starting position to a target along a calculated path. EP 2 910 453 A1 discusses a further autonomous vehicle system, in which the vehicle follows markings inside a car park or a parking area so as to enable the vehicle to be parked. EP 2 136 275 A1 discusses a system that automatically guides a vehicle along a path, and does so within an area.

(5) However, these systems are incomplete as the driver either always retains responsibility for the vehicle or else the system only takes over relatively simple control operations for the vehicle (that is to say acts as an active navigation system). Furthermore, these documents deal with typical applications for passenger vehicles. For utility vehicles, searching for a parking space is not as important as it is for passenger vehicles. Instead of this, it is often important for utility vehicles to execute specific actions on the vehicle, such as for example loading or unloading, maintenance,

refueling, cleaning the vehicle or else navigating other transport arrangement (for example driving onto a train). The cited documents do not disclose any solutions for these cases of application.

(6) There is therefore a need for further automated or semi-automated operations that are able to be executed within areas without interaction with a driver.

SUMMARY OF THE INVENTION

(7) At least some of the abovementioned problems may be solved by a system for operating a driverless utility vehicle as described herein, an associated method as described herein and a management system as described herein. The further descriptions herein define further advantageous embodiments.

(8) The present invention relates to a system for operating a driverless utility vehicle in a restricted area. The restricted area comprises an entrance, an exit and a predetermined target point. The system comprises a transfer module, a movement module and a performance module. The transfer module is configured to transfer control of the utility vehicle at the entrance from the driver of the utility vehicle to the system, and at the exit from the system to the driver of the utility vehicle. The movement module is configured to move the utility vehicle from the entrance to the predetermined target point and from the predetermined target point to the exit autonomously, in particular in a driverless manner. The performance module is configured to perform an action on the utility vehicle autonomously while the utility vehicle is situated at the predetermined target point, so as to change a state of the utility vehicle by way of the action.

(9) A restricted area in the context of the present invention should be understood to mean any area in which regulations are present to the extent that for example access monitoring (for example for individuals) is performed and there is a restriction with regard to vehicle movements. A restricted area is thus intended to comprise in particular areas such as for example loading yards, loading and unloading stations, ferries, aircraft, ships, rail transport arrangement and more. The entrance and the exit may optionally be situated at one and the same location or constitute only different roadways of an entrance.

(10) The term “autonomous” should be understood such that the activity characterized thereby is able to be executed without human input. In particular, the execution does not require any individual to be present—not even for monitoring purposes. Furthermore, the term “operate” should be understood broadly and should not be understood to mean only a movement of the utility vehicle, but should rather also encompass an actuation of any desired actuators of the utility vehicle. This includes for example a height adjustment (for example of a loading surface), actuation of tailgates, coupling/decoupling of a trailer, actuation of a braking device, starting/turning off of machines, etc.

(11) It is understood that the modules defined in the context of the present invention may in particular also be implemented by software in a control unit of the utility vehicle or of another system, such that the control unit of the utility vehicle is able to provide the described functions.

(12) The performance module may optionally be configured to perform the action autonomously, that is to say without interaction with the driver of the utility vehicle or another individual.

(13) The performance module is optionally furthermore configured to change at least one of the following states of the utility vehicle by way of the action: an energy charge state, a loading state, coupling or decoupling of vehicle parts, a global position of the vehicle, soiling of the outer surface of the vehicle, a service state of the vehicle or of one of its components.

(14) The movement module and/or the performance module is/are optionally configured to move the utility vehicle autonomously and/or to perform the action autonomously only when the driver has left the utility vehicle (is no longer in the driver's cabin).

(15) The utility vehicle may have a sensor unit and vehicle actuators for operating the utility vehicle. In this case, the movement module may be configured to obtain sensor data from the sensor unit and to drive the vehicle actuators on the basis of the sensor data in order to operate the utility vehicle autonomously.

- (16) The sensor unit may for example have surroundings sensors and the movement module may optionally be configured to recognize obstacles during the autonomous movement using the surroundings sensors and, on the basis of this, to change a movement path from the entrance to the predetermined target point or from the predetermined target point to the exit, to use an alternative path or to interrupt a movement.
- (17) The system may optionally have an assistance module that is configured to request support from an operator of the system when the movement module is not able to take an alternative path or interrupt the movement.
- (18) The sensor unit may comprise a position sensor and a sensor for determining the orientation of the utility vehicle, and the movement module may optionally be configured to park the utility vehicle at the predetermined target point at a defined orientation.
- (19) The restricted area may be able to be controlled by a management system, and the management system may be configured to plan paths for vehicles in the restricted area. In this case, the system may optionally have a communication interface that is configured to produce a connection with the management system of the restricted area so as to obtain data for a planned path in the restricted area. The management system may be accessed in this case in particular via a radio module, which is already situated in the vehicle or is configured as part of the communication interface.
- (20) The movement module is optionally configured to move the utility vehicle autonomously along the planned path to the further target points within the restricted area, so as to execute further actions at the further target points or to temporarily park the utility vehicle. The communication interface may furthermore be configured to obtain information about actions at the further target points.
- (21) The movement module may furthermore optionally be configured to determine target points for the utility vehicle on the basis of information depending on actions that the driver of the utility vehicle or that a driver of other vehicles have performed previously. By way of example, the utility vehicle may be autonomously controlled such that it follows a route (including potential waiting points) that another vehicle has taken previously or else the driver himself has driven with the utility vehicle (for example for learning purposes).
- (22) The system optionally comprises an information module that is configured to show and/or acoustically indicate the autonomously executed actions in the restricted area, such that surroundings of the utility vehicle obtain information regarding the autonomously executed actions.
- (23) The present invention also relates to a management system having a communication interface and a control unit. The communication interface is configured to produce a communication connection with a system as was described above. The control unit is configured to perform planning and coordination for a self-driving utility vehicle in the restricted area.
- (24) The present invention also relates to a method for operating a driverless utility vehicle in said restricted area. The method comprises the following steps: transferring control of the utility vehicle at the entrance from the driver of the utility vehicle to the system, and at the exit from the system to the driver of the utility vehicle; autonomously moving the utility vehicle from the entrance to the predetermined target point and from the predetermined target point to the exit; and autonomously performing an action on the utility vehicle while the utility vehicle is situated at the predetermined target point, so as to change a state of the utility vehicle by way of the action.
- (25) This method may likewise be implemented or stored in the form of instructions in software or on a computer program product, wherein stored instructions are capable of executing the steps according to the method when the method runs on a processor (for example one of the vehicle control units). Therefore, the present invention likewise relates to a computer program product having software code (software instructions) stored thereon that is configured to execute one of the methods described above when the software code is executed by a processing unit. The processing unit may be any form of computer or control unit that has a corresponding microprocessor that is

able to execute a software code.

(26) At least some of the abovementioned technical problems are therefore solved by a system according to exemplary embodiments in that the driver transfers the utility vehicle to the system, such that the driverless utility vehicle is operated by a system in the restricted area. Working hours (driving hours) of the driver are thus saved, and the utility vehicle is used efficiently.

(27) In comparison with known systems, exemplary embodiments offer the following advantages:

(28) The risks of accidents are significantly reduced, as the driver is replaced by the system, which autonomously takes over the vehicle within the restricted area and at the same time executes a predetermined action on the vehicle. The driver is completely decoupled from interaction with the utility vehicle. The driver in particular does not need to monitor the loading or unloading or the maintenance or refueling of the vehicle. These activities are executed automatically by the system, such that the driver does not need to be available in the vehicle. Since the autonomous operations are executed in a restricted area and individuals are therefore able to be refused entry or only have limited access, exemplary embodiments likewise ensure a high degree of safety.

(29) The exemplary embodiments of the present invention will be better understood from the following detailed description and the attached drawings of the various exemplary embodiments, which however are not intended to be understood such that they restrict the disclosure to the specific embodiments, but rather serve merely for the purposes of explanation and understanding.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows an exemplary embodiment for a system for operating a utility vehicle without a driver in a restricted area.

(2) FIG. 2 shows a vehicle as may be used in exemplary embodiments.

(3) FIG. 3 shows an exemplary restricted area.

(4) FIG. 4 illustrates possible maneuvers that need to be executed by a utility vehicle in the restricted area.

(5) FIG. 5 shows a flow diagram for a method that is executed by the system according to exemplary embodiments.

DETAILED DESCRIPTION

(6) FIG. 1 shows an exemplary embodiment for a system **100** that is suitable for operating a utility vehicle in a restricted area, without a driver having to be present in the utility vehicle. The restricted area comprises an entrance, an exit and has a predetermined target point. The system **100** comprises a transfer module **110**, a movement module **120** and a performance module **130**. The transfer module **110** is configured to transfer control of the utility vehicle, and to do so at the entrance: from the driver of the utility vehicle to the system, and at the exit: from the system to the driver of the utility vehicle. The movement module **120** is configured to move the utility vehicle from the entrance to the predetermined target point **201** and from the predetermined target point to the exit autonomously. The performance module **130** is configured to perform an action on the utility vehicle autonomously while the utility vehicle is situated at the predetermined target point. This action changes a state of the utility vehicle.

(7) The system is therefore capable of moving the utility vehicle within the area and performing position changes autonomously. It is possible in particular for one or more target points within the area to be traveled through in a particular order. At the predetermined target point, a management system of the restricted area may make particular changes to the vehicle during the stop. A typical action is for example loading or unloading, but also cleaning or maintenance of the utility vehicle or refueling or recharging of batteries (or changing of a general energy charge state) or any desired combination thereof.

(8) The system **100** may be fully or partly implemented by software that is installed in a control unit of the utility vehicle or in another system in order to execute the described functions. The other system may in particular be outside the utility vehicle (for example be part of the management system).

(9) FIG. 2 shows an exemplary embodiment for a vehicle **10**, in particular a utility vehicle, which is able to be used so as to be operated autonomously within the restricted area. The utility vehicle **10** comprises the system **100**, which may be installed for example inside a controller of the vehicle **10** or be accommodated in the vehicle **10** as separate hardware.

(10) The utility vehicle **10** furthermore comprises at least one sensor unit **12** that is configured to detect surroundings of the vehicle **10** (for example to determine distances to other objects or to detect changes in movement or changes in speed). The utility vehicle **10** furthermore comprises a multiplicity of vehicle actuators **14**, which are configured for example to control the vehicle **10** along a path. These include in particular steering and brake actuators that steer or brake the vehicle **10**. These also include vehicle actuators **14** that are able to accelerate the vehicle **10**. The vehicle optionally comprises a dedicated communication module **16** that is able to produce a radio connection with an external network. The system **100** may be configured for example to use the communication module **16** of the vehicle **10** to obtain data from the external network or to transmit said data thereto. The communication module **16** may furthermore be used to connect the utility vehicle **10** to a management system of the restricted area and to obtain data therefrom for a path to be traveled.

(11) The utility vehicle **10** may comprise for example a bus, a lorry, a traction machine for a trailer or a combination of a plurality of vehicle parts (including a trailer).

(12) FIG. 3 shows an exemplary restricted area that has an entrance **21**. Access monitoring may be provided at the entrance **21** such that it is ensured that only the utility vehicle **10** or vehicles that meet the requirements of the restricted area are active in the restricted area. The utility vehicle **10** may for example be a vehicle, as is shown in FIG. 2.

(13) The restricted area that is shown may be managed by a management system. The management system may be configured for example to determine a path **200** for a utility vehicle **10** that registers appropriately at the entrance **21**, the utility vehicle **10** then being able to travel this path autonomously. This management system may in particular define the points and the order in which the points should be traveled through. The management system may furthermore provide a path specification for the series of waypoints (for example which turns should be made at which positions). It is optionally likewise possible for the management system to supply data that should be considered to be constraints for planning a path along the series of waypoints.

(14) The path **200** leads for example past various buildings **310**, **320**, **330**, which are able to be detected for example by the sensor unit **12** of the utility vehicle **10** using a surroundings sensor system. The path **200** then leads for example to two predetermined target points **201**, **202**, which are docking stations at a warehouse **310**, for example, where the utility vehicle **10** docks and performs loading and unloading activities, for example. Potential further target points are the stations “1”, “2”, . . . , “8”, at which further actions may optionally be performed (for example depending on the type of utility vehicle). It is understood in the context of the present invention that even more intermediate points may be defined at which no actions are executed on the vehicle **10**, but rather the vehicle waits in a parking position, for example. The parking may in this case be only temporary or else for a relatively long time interval.

(15) The management system of the restricted area may furthermore plan and/or control the actions to be executed on the vehicle **10** at the various target points **201**, **202** or the further stations within the area. To this end, the movement module **120** may obtain planning data from the management system and navigate the vehicle **10** autonomously to the various target points. After all of the planned actions have finished, the movement module **120** may navigate the vehicle **10** autonomously to the exit **22**. The exit **22** may be the same point as the entrance **21**, at which the

driver of the vehicle **10**, using the transfer module **110**, is able to take back control of the vehicle **10** and drives the vehicle **10** out of the restricted area through manual driving.

(16) The system **100** may likewise comprise interfaces to the vehicle sensor unit **12** and to the vehicle actuators **14** in order to establish a state of the vehicle **10** and of the surroundings (for example detect the buildings **320**, **330**, **310** or a relative orientation with respect thereto) and, on the basis thereof, to control the vehicle state, and do so without assistance from the driver being necessary for this purpose. This may be for example an interface to the management system of the restricted area, such that the management system retains access to the vehicle **10** is able to access components of the vehicle. This interface may allow wireless communication, for example.

(17) The restricted area that is shown may be managed by an operator, for example. The operator may in particular be or comprise the management system that is configured to ensure automated operation for the restricted area. The management system may however optionally likewise be operated by an individual or at least partly controlled or monitored so as to ensure the safety of the restricted area.

(18) According to further exemplary embodiments, it is likewise possible to define a journey between the target points such that a path that has been taken previously is repeated. By way of example, a path may have been originally traveled manually by the driver and movement data may have been captured. The movement module **120** is then configured for example, on the basis of the movement data, to allow the utility vehicle **10** to travel autonomously along the recorded path.

(19) The restricted area that is shown constitutes just an example. The system **100** is however in particular also able to be used in areas where high accuracy in terms of the navigation of the vehicle **10** is necessary within a short time, and in which damage often occurs when utility vehicles **10** are operated manually. Especially in such areas, the system **100** ensures safe operation of the utility vehicle **10**. By way of example, these areas are other transport vehicles (a ferry, a train, an aircraft, a road train, etc.) on which the utility vehicle **10** is intended to be parked with high precision. In this case, the movement of the utility vehicle **10** (change in its global position together with the transport vehicle) is the action that is performed in the restricted area. The utility vehicle **10** may thus also adopt a fixed point in relation to the area during the action, and the restricted area (for example the interior of an aircraft) moves.

(20) FIG. 4 illustrates possible maneuvers that need to be executed by a utility vehicle **10** on/in the restricted area. First of all, the vehicle **10** registers at the entrance **21** with the operator/management system **207**. In this case, the vehicle **10** may be transferred to the system **100**, which registers with the management system for the restricted area. After the utility vehicle **10** has been transferred, the driver **11** may leave the utility vehicle **10**. The utility vehicle **10** may then be operated autonomously within the restricted area.

(21) By way of example, FIG. 4 shows a situation in which the utility vehicle **10** may potentially collide with a forklift truck **340** at a position **350**. This situation may be handled by the system **100** in that the utility vehicle **10** initially waits in a waiting position until the forklift truck **340** has finished its path **345**. The system **100** may then move the utility vehicle **10** further along the path **200** to a turning position **203**. At the turning position **203**, the movement direction of the utility vehicle **10** is reversed, and reverse docking at a docking station **1** of a building is performed (position **204**). At this position **204**, the utility vehicle **10** may be loaded, for example.

(22) The system **100** may then obtain corresponding information from the management system as to how long the utility vehicle **10** should wait at the position **204**. Following authorization by the management system **200**, the system **100** may continue the movement from the position **204**. This may be performed for example in that the management system confirms to the system **100** that the loading activity at the docking station **204** has finished and the movement is able to be continued. The system **100** may then autonomously further guide the utility vehicle **10** from the docking station **204** to a parking position **205** (or end point) at the exit **22**. At the exit **22**, the driver is able to receive the utility vehicle **10** again and drive manually.

(23) FIG. 4 thus illustrates a typical reversal of movement of the utility vehicle **10** that constitutes a high accident risk in complex loading yards when the utility vehicle **10** is operated manually and the driver does not have accurate knowledge of the loading yard. For this reason, the system **100** is able to be used advantageously in particular for such situations.

(24) It is understood that it may be necessary to determine and to monitor the position and orientation of the utility vehicle **10** as accurately as possible, in order thus to be able to efficiently execute the actions at the points within the area. For this purpose, the management system of the restricted area may require for example an accurate position determination by the utility vehicle **10**, that is to say, for a successful transfer, it may be necessary for the utility vehicle **10** to have sufficiently accurate and reliable sensors for position determination (for example rangefinders, radar systems, cameras, ultrasound sensors, etc.).

(25) FIG. 5 shows a flow diagram for a method that is able to be executed for example by the system **100**, wherein the system in this case is at least partly integrated into the management system for the restricted area.

(26) In step **410**, the restricted area is entered. In step **420**, the utility vehicle **10** is taken over by the operator (or the management system) that operates the restricted area using the management system. In step **430**, the management system defines target points that are to be traveled through by the utility vehicle **10**. In this case, the management system may in particular transfer further information about the path **200** to the defined target points.

(27) In step **440**, it is established whether a target point that is the first one or the next one to be traveled through is ready to accept the utility vehicle **10**. If so, in step **450**, the path to the next target point is generated and, in step **460**, the next target point is reached and the action is executed on the utility vehicle **10**. When this has been performed, in step **470**, it is initially queried whether the target point is a last target point. If so, the management system generates a path to the exit in step **480**. In step **490**, the driver receives the utility vehicle **10** and takes over control of the utility vehicle **10**. And, in step **495**, the driver manually drives the vehicle **10** out of the restricted area.

(28) If, at the query **440**, the first or next target point should not yet be ready to accept the utility vehicle **10**, in step **442**, the utility vehicle **10** is guided to a parking position, for which purpose a corresponding path to the parking position is generated. In step **444**, the vehicle then travels to the parking position and waits there until there is confirmation that the next target point may be traveled to. If this is the case, the utility vehicle **10** follows the previously determined path (step **450**).

(29) If the system **100** establishes that the target point that was traveled through in step **460** and the action that was executed there is not yet the last target point in the restricted area, there is a new query as to whether the next target point is ready to accept the utility vehicle **10**. This is performed in step **440**. If so, the method continues with steps **450** and **460**, and if this is not possible, steps **442** and **444** follow, as described above.

(30) The method or at least parts thereof may likewise be computer-implemented, that is to say it may be implemented through instructions that are stored on a storage medium and are capable of executing the steps of the method when it runs on a processor. The instructions typically comprise one or more instructions that are able to be stored in various ways on various media in or peripherally with respect to a control unit (having a processor) that, when they are read and executed by the control unit, prompt the control unit to execute functions, functionalities and operations that are necessary to execute a method according to the present invention.

(31) Further exemplary embodiments of the present invention relate to the following subjects:

(32) A system **100** for a utility vehicle (or a vehicle combination) **10**, which is operated at least partly in an area where the entrance **21** and exit **22** are monitored, so as to travel from an entry of the area at least to a predetermined target point at which a predetermined action is executed on the vehicle **10** or the vehicle combination. Furthermore, this position is left and the exit from the area is controlled. This system is characterized in that the action is performed on the vehicle or the vehicle

combination **10** autonomously within the area, without the driver being situated in the vehicle or the vehicle combination **10**.

(33) The present invention also relates to a system in which responsibility for the vehicle is transferred from the driver to a management system of the area.

(34) Further exemplary embodiments relate to a system **100** in which the predetermined action comprises any desired action that results in a change of at least one vehicle property.

(35) Further exemplary embodiments relate to a system **100** in which the vehicle property is an energy storage level, a loading level, vehicle coupling or decoupling, a global position of the vehicle **10**, soiling on the outer surface or a maintenance state with respect to an additional component of the vehicle or the vehicle combination itself.

(36) Further exemplary embodiments relate to a system **100** in which, between leaving the entrance **21** and reaching the exit **22**, any desired number of target points may be defined that are controlled by the utility vehicle **10**, with or without specific actions having to be executed there.

(37) Further exemplary embodiments relate to a system **100** in which the path **200** of the vehicle or the vehicle combination **10**, between the entrance **21** and the exit **22**, comprises points within the area to be traveled through and that are determined by the supervisor of the area or are determined by the vehicle **10** on the basis of information provided by the supervisor of the area or based on data from previously performed manual operations by the driver of the vehicle **10** or a driver of another vehicle **10**.

(38) The supervisor of the area may optionally define the points to be traveled through and stipulate the orientation of the utility vehicle upon approach.

(39) The system optionally has interfaces to the driver, vehicle sensors, to vehicle actuators and to the supervisor of the area. The interfaces may be part of a mobile device, or have a display or input arrangement.

(40) The features of the invention that are disclosed in the description, the claims and the figures may be essential to implementing the invention both individually and in any desired combination.

THE LIST OF REFERENCE SIGNS IS AS FOLLOWS

(41) **10** utility vehicle/vehicle combination **12** sensor unit **14** vehicle actuators **16** communication module **21** entrance **22** exit **100** system **110** transfer module **120** movement module **130** performance module **201**, **202** predetermined target points **203** reversing point **204** loading point **205** parking position **207** operator/management system **340** exemplary forklift truck **350** potential intersection **345** path of the forklift truck

Claims

1. A system for operating a driverless utility vehicle in a restricted area, for installation in a control unit of the utility vehicle, comprising: a transfer module for transferring control of the utility vehicle at an entrance from the driver of the utility vehicle to the system, and at an exit from the system to the driver of the utility vehicle, wherein the restricted area has the entrance, the exit and a predetermined target point, wherein the utility vehicle has a sensor unit and vehicle actuators for operating the utility vehicle and the sensor unit includes surroundings sensors; a movement module for autonomously moving the utility vehicle along a path from the entrance to the predetermined target point and from the predetermined target point to the exit, wherein the movement module is configured to obtain sensor data from the sensor unit and to drive the vehicle actuators based on the sensor data to operate the utility vehicle autonomously, and wherein the movement module is configured to recognize obstacles during the autonomous movement using the surroundings sensors, and, based on this, to change a movement path from the entrance to the predetermined target point or from the predetermined target point to the exit, to use an alternative path or to interrupt a movement, wherein the path between the entrance and the exit includes points within the restricted area to be traveled through, the movement module determining the points based on

movement data captured during previously performed manual operations by the driver of the utility vehicle; a performance module for autonomously performing an action on the utility vehicle while the utility vehicle is situated at the predetermined target point, so as to change a state of the utility vehicle by the action; and an assistance module to request support from an available management system or an operator of the management system when the movement module is not able to take an alternative path or interrupt the movement; wherein the transfer module, the movement module, and the performance module are installed in the control unit of the utility vehicle, wherein following authorization by the management system, the utility vehicle is autonomously guided from a docking station to a parking position or end point at the exit, where the driver receives the utility vehicle to drive the utility vehicle manually, the authorization depending on availability of the management system, and wherein the movement module, which is installed in the control unit of the utility vehicle, is configured to move the utility vehicle along the path from the entrance to the predetermined target point and from the predetermined target point to the exit autonomously.

2. The system of claim 1, wherein the performance module is configured to perform the action autonomously, without interaction with the driver of the utility vehicle or another individual.
3. The system of claim 1, wherein the performance module is configured to change at least one of the following states of the utility vehicle by the action: an energy charge state, a loading state, coupling or decoupling of vehicle parts, a global position of the vehicle, soiling of the outer surface of the vehicle, a service state of the vehicle or of one of its components.
4. The system of claim 1, wherein the movement module and/or the performance module are configured to move the utility vehicle autonomously and to perform the action autonomously only when the driver has left the utility vehicle.
5. The system of claim 1, wherein the sensor unit includes a position sensor and a sensor for determining the orientation of the utility vehicle, and wherein the movement module is configured to park the utility vehicle at the predetermined target point at a defined orientation.
6. The system of claim 1, wherein the restricted area is controllable by the management system, and the management system is configured to plan paths for vehicles in the restricted area, further comprising: a communication interface to establish a connection with the management system of the restricted area to obtain data for a planned path in the restricted area.
7. The system of claim 6, wherein further target points lie along the planned path, and wherein the movement module is configured to move the utility vehicle autonomously along the planned path to the further target points within the restricted area, so as to execute further actions at the further target points or to temporarily park the utility vehicle, and wherein the communication interface is configured to obtain information about actions at the further target points.
8. The system of claim 1, further comprising: an information module to visualize and/or acoustically indicate the autonomously executed actions in the restricted area, such that surroundings of the utility vehicle obtain information regarding the autonomously executed actions.
9. A management system, comprising: a communication interface to establish a communication connection with a system for operating a driverless utility vehicle in a restricted area, for installation in a control unit of the utility vehicle, wherein the system includes: a transfer module for transferring control of the utility vehicle at an entrance from the driver of the utility vehicle to the system, and at an exit from the system to the driver of the utility vehicle, wherein the restricted area has the entrance, the exit and a predetermined target point, wherein the utility vehicle has a sensor unit and vehicle actuators for operating the utility vehicle and the sensor unit includes surroundings sensors; a movement module for autonomously moving the utility vehicle along a path from the entrance to the predetermined target point and from the predetermined target point to the exit, wherein the movement module is configured to obtain sensor data from the sensor unit and to drive the vehicle actuators based on the sensor data to operate the utility vehicle autonomously, and wherein the movement module is configured to recognize obstacles during the autonomous movement using the surroundings sensors, and, based on this, to change a movement path from the

entrance to the predetermined target point or from the predetermined target point to the exit, to use an alternative path or to interrupt a movement; a performance module for autonomously performing an action on the utility vehicle while the utility vehicle is situated at the predetermined target point, so as to change a state of the utility vehicle by the action; and an assistance module to request support from the management system or an operator of the management system when the movement module is not able to take an alternative path or interrupt the movement; wherein the transfer module, the movement module, and the performance module are installed in the control unit of the utility vehicle, wherein following authorization by the management system, the utility vehicle is autonomously guided from a docking station to a parking position or end point at the exit, where the driver receives the utility vehicle to drive the utility vehicle manually, and wherein the movement module, which is installed in the control unit of the utility vehicle, is configured to autonomously move the utility vehicle along the path from the entrance to the predetermined target point and from the predetermined target point to the exit autonomously, wherein the path between the entrance and the exit includes points within the restricted area to be traveled through, the movement module determining the points based on movement data captured during previously performed manual operations by the driver of the utility vehicle; and a control unit to perform planning and coordination for an autonomously driven utility vehicle in the restricted area.

10. A method for operating a driverless utility vehicle in a restricted area, the method comprising: transferring control of the utility vehicle, by a transfer module in the utility vehicle, at an entrance from the driver of the utility vehicle to the system, and at an exit from the system to the driver of the utility vehicle, wherein the restricted area has the entrance, the exit and a predetermined target point, wherein the utility vehicle has a sensor unit and vehicle actuators for operating the utility vehicle and the sensor unit includes surroundings sensors; autonomously moving the utility vehicle, by a movement module in the utility vehicle, along a path from the entrance to the predetermined target point and from the predetermined target point to the exit, wherein the path between the entrance and the exit includes points within the restricted area to be traveled through, the movement module determining the points based on movement data captured during previously performed manual operations by the driver of the utility vehicle; autonomously performing, by a performance module in the utility vehicle, an action on the utility vehicle while the utility vehicle is situated at the predetermined target point, so as to change a state of the utility vehicle by the action; obtaining of sensor data from the sensor unit and driving the vehicle actuators based on the sensor data to operate the utility vehicle autonomously; recognizing obstacles during the autonomous movement using the surroundings sensors and, based on this, changing a movement path from the entrance to the predetermined target point or from the predetermined target point to the exit, to use an alternative path or to interrupt a movement; and requesting, by an assistance module in the utility vehicle, of support from an available management system or an operator of the management system when the movement module is not able to take an alternative path or interrupt the movement; wherein the transfer module, the movement module, and the performance module are installed in the control unit of the utility vehicle, wherein following authorization by the management system, the utility vehicle is autonomously guided from a docking station to a parking position or end point at the exit, where the driver receives the utility vehicle to drive the utility vehicle manually, the authorization depending on availability of the management system, wherein the steps are performed by the control unit in the utility vehicle, and wherein the movement module, which is installed in the control unit of the utility vehicle, is configured to move the utility vehicle along the path from the entrance to the predetermined target point and from the predetermined target point to the exit autonomously.

11. A non-transitory computer readable medium having a computer program, which is executable by a processor of a control unit, comprising: a program code arrangement having program code for operating a driverless utility vehicle in a restricted area, by performing the following: transferring control of the utility vehicle at an entrance from the driver of the utility vehicle to the system, and

at an exit from the system to the driver of the utility vehicle, wherein the restricted area has the entrance, the exit and a predetermined target point, wherein the utility vehicle has a sensor unit and vehicle actuators for operating the utility vehicle and the sensor unit includes surroundings sensors; autonomously moving the utility vehicle, by a movement module in the utility vehicle, along a path from the entrance to the predetermined target point and from the predetermined target point to the exit, wherein the path between the entrance and the exit includes points within the restricted area to be traveled through, the movement module determining the points based on movement data captured during previously performed manual operations by the driver of the utility vehicle; autonomously performing, by a performance module in the utility vehicle, an action on the utility vehicle while the utility vehicle is situated at the predetermined target point, so as to change a state of the utility vehicle by the action; obtaining of sensor data from the sensor unit and driving the vehicle actuators based on the sensor data to operate the utility vehicle autonomously; recognizing obstacles during the autonomous movement using the surroundings sensors and, based on this, changing a movement path from the entrance to the predetermined target point or from the predetermined target point to the exit, to use an alternative path or to interrupt a movement; and requesting, by an assistance module in the utility vehicle, of support from an available management system or an operator of the management system when the movement module is not able to take an alternative path or interrupt the movement; wherein the transfer module, the movement module, and the performance module are installed in the control unit of the utility vehicle, wherein following authorization by the management system, the utility vehicle is autonomously guided from a docking station to a parking position or end point at the exit, where the driver receives the utility vehicle to drive the utility vehicle manually, the authorization depending on availability of the management system, wherein the steps are performed by the control unit in the utility vehicle, and wherein the movement module, which is installed in the control unit of the utility vehicle, is configured to move the utility vehicle along the path from the entrance to the predetermined target point and from the predetermined target point to the exit autonomously.
